

UNNORMALIZED FORM (UNF)

LIBRARY(

Author_ID, Author_Name,
Book_ID, Title, Category, Price,
Pub_ID, Publisher_Name,
Publisher_Address,
Copy_ID, Status, Shelf_No, Available,
Member_ID, Member_Name,
Member_Address,
Memb_Date, Expiry_Date,
Issue_ID, Issue_Date, Due_Date,
Return_Date, Fine
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FIRST NORMAL FORM (1NF)

AUTHOR(

Author_ID PK,
Name
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PUBLISHER(

Pub_ID PK,
Name,
Address
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BOOK(
Book_ID PK,
Title,
Category,
Price,
Pub_ID FK
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Author_ID PK/FK,
Book_ID PK/FK
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Memb_Date,
Expiry_Date
)

ISSUE(
Issue_ID PK,
Issue_Date,
Due_Date,
Return_Date,
Fine
)

SECOND NORMAL FORM (2NF)

AUTHOR(
Author_ID PK,
Name
)

PUBLISHER(
Pub_ID PK,
Name,
Address
)

BOOK(
Book_ID PK,
Title,
Category,
Price,
Pub_ID FK
)

WRITES(
Author_ID PK/FK,
Book_ID PK/FK
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Copy_ID PK,
Status,
Shelf_No,
Available,
Book_ID FK
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MEMBER(
Member_ID PK,
Name,
Address,
Memb_Date,
Expiry_Date
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ISSUE(
Issue_ID PK,
Issue_Date,
Due_Date,
Return_Date,
Fine
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THIRD NORMAL FORM (3NF)

AUTHOR(

Author_ID PK,
Name
)

PUBLISHER(

Pub_ID PK,
Name,
Address
)

BOOK(

Book_ID PK,
Title,
Category,
Price,
Pub_ID FK
)

WRITES(

Author_ID PK/FK,
Book_ID PK/FK
)

COPY(

Copy_ID PK,
Status,
Shelf_No,
Available,
Book_ID FK
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MEMBER(

Member_ID PK,
Name,
Address,
Memb_Date,
Expiry_Date
)

ISSUE(

Issue_ID PK,
Issue_Date,
Due_Date,
Return_Date,
Fine,
Member_ID FK,
Copy_ID FK
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